

SUPPLIER SPECIFICATION MANUAL

Component: ADAS Radar Sensor Module

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Classification: Automotive Grade / ASIL-B

Revision Date: Rev 3.4

1. Scope

This document specifies the technical characteristics, interface protocols, and environmental survivability requirements for the Continental ADAS Radar Sensor, used in adaptive cruise control, automatic emergency braking, and blind-spot detection systems.

2. Functional Overview

- 77 GHz FMCW radar operation for long- and mid-range object detection
- Detection range up to 250m (long-range mode)
- Simultaneous tracking of up to 64 objects
- Real-time relative velocity and angular position calculation
- Sensor fusion-ready output format for ADAS domain controller integration

3. Electrical Interface

- Nominal supply voltage: **12V DC** (vehicle electrical system)
- Operating voltage range: 9V–16V DC
- Reverse polarity protection integrated
- Max power consumption: 8.5W (active scanning mode)

4. Communication Interfaces

The sensor utilizes high-speed CAN FD communication protocols for object list transmission and diagnostic messaging.

- CAN FD data rate: up to 5 Mbit/s (data phase), 500 kbit/s (arbitration phase)
- Supports ISO 15765-4 (CAN FD diagnostic communication)
- Backward-compatible CAN 2.0B mode available via configuration flag

5. Environmental Limits

- Operating temperature range: **-40°C to +125°C**
- Storage temperature range: -40°C to +125°C
- IP rating: IP69K (housing, mated connectors)
- Vibration resistance per ISO 16750-3
- EMC compliance per CISPR 25 Class 5

6. Technical Specifications Table

Parameter	Value	Unit	Notes
Operating Frequency	76–81	GHz	FMCW radar band
Detection Range (Long)	250	m	±0.5m accuracy
Field of View	120	degrees	Combined long/short mode
Supply Voltage	12	V DC	9–16V range
Power Consumption	8.5	W	Active mode
Communication Protocol	CAN FD	—	ISO 15765-4
Data Rate (CAN FD)	5	Mbit/s	Data phase
Min Operating Temperature	-40	°C	Cold environment rated
Max Operating Temperature	+125	°C	Engine-bay rated
Ingress Protection	IP69K	—	High-pressure wash rated
Functional Safety Rating	ASIL-B	—	ISO 26262

7. Compliance & Certification

- ISO 26262:2018 (ASIL-B)
- ISO 16750 (Environmental conditions)
- CISPR 25 (EMC)
- IATF 16949 manufacturing certification